

Awesom Living — AI Features Developer Quick Reference

Architecture: Backend gathers data → builds prompt → Claude Sonnet returns text → Backend stores result.

Ground Rules: (1) Backend decides all facts, AI only writes sentences. (2) AI never makes changes to settings/data. (3) One AI on/off toggle per Home. (4) Correlation only — never say one thing *caused* another.

Shared Source Collections (Already in MongoDB)

sleep_sessions

```
{ "resident_id": "ObjectId", "platform_date": "2026-10-14", "status": "complete",
  "duration_minutes": 460, "stages": { "light": 210, "deep": 120, "rem": 90, "awake": 40 },
  "avg_heart_rate": 64, "avg_respiration": 15, "movement_event_count": 12,
  "snoring_detected": true, "toss_turn_count": 8 }
```

platform_date = calendar date person went to bed (2 AM on Oct 15 → platform_date Oct 14).

bp_readings

```
{ "home_id": "ObjectId", "resident_id": "ObjectId", "systolic": 142, "diastolic": 88,
  "pulse": 72, "taken_at": "2026-10-15T08:30:00Z", "received_at": "...",
  "mapped_at": "...", "mapped_by": "account_holder_id" }
```

Readings arrive unclaimed. Family member claims a reading to a Resident via the app.

room_occupancy_state / room_occupancy_events

```
// state: { "resident_id": "ObjectId", "room_id": "ObjectId", "occupied": true, "since": "..."}
// events: { "resident_id": "ObjectId", "room_id": "ObjectId", "event": "entered", "timestamp": "..."} 
```

alerts

```
{ "resident_id": "ObjectId", "home_id": "ObjectId", "alert_type": "no_motion",
  "triggered_at": "...", "resolved_at": "...", "status": "resolved",
  "resolution_reason": "confirmed_concern", "context": { "room": "bathroom", "threshold_minutes": 30 } }
```

resolution_reason: false_positive | confirmed_concern — needed by Feature 6.

room_alert_settings

```
{ "resident_id": "ObjectId", "room_id": "ObjectId",
  "long_stay_threshold_minutes": 45, "no_motion_threshold_minutes": 30 }
```

monthly_summary

```
{ "resident_id": "ObjectId", "month": "2026-10", "avg_sleep_duration_minutes": 440,
  "avg_heart_rate": 65, "avg_bp_systolic": 128, "avg_bp_diastolic": 82,
  "bp_reading_count": 6, "alert_counts_by_type": { "no_motion": 1 }, "total_alerts": 1 }
```

Pre-computed per Resident per calendar month.

dashboard_section_views

```
{ "account_holder_id": "ObjectId", "home_id": "ObjectId",
  "section": "motion_tracking", "viewed_at": "2026-10-15T21:00:00Z" }
```

Feature-by-Feature Implementation

Feature 1 — Daily Narrative

What:

Evening card on Dashboard. One headline per Home, one sentence + status pill + stat chips per Resident.

Source Collections:

sleep_sessions (by platform_date), bp_readings (if claimed that day), room_occupancy_state/events, alerts, device status (Home-level).

New Collections:

daily_narratives (id, home_id, platform_date, headline, status), resident_narratives (id, daily_narrative_id, resident_id, narrative_text, resident_status [ok|attention], flags, stats).

API:

Nightly cron job generates. One Claude call per Home (all Residents at once).

Notes:

Headline is templated (not AI). Backend pre-decides flags[] per Resident (short_sleep, bp_out_of_range, occupancy_alert, emergency_pressed). Fallback if AI fails: plain numbers.

Feature 2 — Ask Abhi on Alerts

What:

Button on every alert card. Explains this one alert + shows 30-day history pattern for same alert_type + Resident.

Source Collections:

alerts (filter by resident_id + alert_type, last 30 days).

New Collections:

alert_explanations (id, alert_id, explanation, historical_analysis, overall_note, generated_at). Cached per alert.

API:

POST /api/alerts/{alert_id}/explanation

Notes:

Backend pre-computes: total_count, most_common_time, trend (going_up|stable|going_down). AI gets aggregated numbers, never raw alert list. Fallback: show plain stats + retry button.

Feature 3 — Ask Abhi Chatbot

What:

FAB button on Dashboard → chat screen. User asks free-text questions about their Residents/Homes.

Source Collections:

All collections — accessed via fixed tool list: get_sleep_summary (sleep_sessions), get_bp_readings (bp_readings), get_occupancy_summary (room_occupancy_state/events), get_door_window_events (door/window log), get_alerts (alerts). AI picks tool + params, Backend executes safe query.

New Collections:

conversations (chat history per Home).

API:

POST /api/homes/{home_id}/chat, GET /api/homes/{home_id}/chat/recent

Notes:

SECURITY: AI never writes DB queries directly. Backend validates ownership. Date ranges capped 12 months. Resident resolution: match real names, clarify vague references.

Feature 4 — Monthly Review & Compare Months

What:

Two screens per Resident. Monthly Review card: headline + 4 stat tiles (sleep, resting_hr, bp, alerts) with better/same/watch. Compare Months: same 4 metrics across last 5 months.

Source Collections:

monthly_summary (pre-computed from sleep_sessions, bp_readings, alerts).

New Collections:

None — monthly_summary already exists.

API:

GET /api/residents/{resident_id}/monthly-review?month=2026-10, GET /api/residents/{resident_id}/monthly-trend?metric=sleep&months=5

Notes:

Classification rules (Backend decides): Sleep better if above avg, watch if below. BP same if in range. Alerts better if count down. Edge: no BP → return nothing (not zero). First month → no comparison_label.

Feature 5 — Doctor Report

What:

Side Menu → "Generate report with Abhi." One-month PDF for a Resident, meant for their doctor.

Source Collections:

monthly_summary (reuse Feature 4 data) + sleep_sessions (movement/snoring/toss fields).

New Collections:

doctor_reports (id, resident_id, period_start, period_end, generated_at, status [generating|ready|failed], findings, pdf_url).

API:

POST /api/residents/{resident_id}/doctor-report

Notes:

Backend ranks findings by severity (bp_out_of_range=3, sleep_decline=2, unresolved_alerts=3). AI writes prose per finding. Backend appends device disclaimer AFTER AI. Correlation language only — never 'caused.'

Feature 6 — Adaptive Threshold Advisor

What:

Weekly suggestion card. Suggests loosening or tightening no_motion/long_stay threshold based on alert history.

Source Collections:

alerts (with resolution_reason), room_alert_settings.

New Collections:

threshold_suggestions (id, resident_id, alert_type, direction [loosen|tighten], current_threshold_minutes, suggested_threshold_minutes, supporting_incident_ids, status [pending|dismissed|applied]).

API:

Weekly cron job. Dashboard shows pending suggestions.

Notes:

Loosen: 5+ false_positive alerts, none confirmed_concern. Tighten: 2+ confirmed_concern where time < current setting. PREREQUISITE: alerts.resolution_reason must exist.

Feature 7 — Personalised Insight Feed

What:

Suggests reordering Dashboard sections based on which detail pages the Account Holder opens most.

Source Collections:

dashboard_section_views (last 7 days, sorted by count).

New Collections:

dashboard_suggestions (id, account_holder_id, home_id, proposed_order, supporting_counts, status [pending|dismissed|applied]).

API:

Periodic check. Dashboard shows pending suggestion.

Notes:

Trigger: top_section by view count NOT already first AND count >= 2x second. PREREQUISITE: Customise Dashboard must support opening with a suggested unsaved order.

Feature 8 — Ask Abhi on Graphs

What:

Button on every graph in the app. Explains that graph in plain words and flags anything noteworthy.

Source Collections:

Depends on graph_type: sleep_sessions (sleep_stage_timeline), bp_readings (bp_trend), monthly_summary (monthly_metric_trend).

New Collections:

graph_explanations (id, graph_type, resident_id, context_hash, simple_explanation, flag_text, generated_at). Cached by context_hash.

API:

POST /api/graphs/explanation

Notes:

Unified analysis shape — one backend function per graph_type returns: summary_stats, baseline_comparison, is_significant, flags. AI called identically for all graph types.

Prerequisites — Build Before Any Feature

#	What	Blocks	Build
1	AI on/off toggle per Home	ALL 8	One toggle in Home settings. Off: numbers only, Ask Abhi buttons hidden
2	alerts.resolution_reason	F2, F6	When closing alert: save false_positive or confirmed_concern
3	sleep_sessions movement/snoring/toss	F5, F8	Verify GLK sends these fields, confirm data types
4	Customise Dashboard suggested-order	F7	Must open with a proposed order not yet saved
5	General door/window event list	F3	Log of gate opens/closes separate from occupancy detection
6	Single alerts table migration	F2, F6	Migrate any alert type still in separate table to unified alerts

New API Endpoints

Endpoint	Method	Feature
/api/alerts/{alert_id}/explanation	POST	2: Ask Abhi on Alerts

/api/homes/{home_id}/chat	POST	3: Chatbot
/api/homes/{home_id}/chat/recent	GET	3: Chatbot
/api/residents/{resident_id}/monthly-review	GET	4: Monthly Review
/api/residents/{resident_id}/monthly-trend	GET	4: Compare Months
/api/residents/{resident_id}/doctor-report	POST	5: Doctor Report
/api/graphs/explanation	POST	8: Graphs

New Collections Summary

Collection	Feature	Purpose
daily_narratives	1: Daily Narrative	One per Home per evening
resident_narratives	1: Daily Narrative	Per-Resident narrative under daily_narratives
alert_explanations	2: Ask Abhi on Alerts	Cached explanation per alert
conversations	3: Chatbot	Chat history per Home
doctor_reports	5: Doctor Report	Generated reports with PDF links
threshold_suggestions	6: Threshold Advisor	Pending/dismissed/applied suggestions
dashboard_suggestions	7: Insight Feed	Dashboard reorder suggestions
graph_explanations	8: Graphs	Cached graph explanations